

FORCE AND TORQUE ANALYSIS

The total weight of the robot is calculated as:

$$W = mg = 4.5 \times 9.81 = 44\text{N}$$

The rolling resistance is:

$$F_{\text{rolling}} = 0.05 \times 44 = 2.2\text{N}$$

The slope force for a 10° inclination is:

$$F_{\text{slope}} = 44 \sin 10^\circ \approx 7.6\text{N}$$

Thus, total force:

$$F_{\text{required}} = 10\text{N}$$

Applying safety factor:

$$F_{\text{design}} = 30\text{N}$$

Torque required:

$$T = 30 \times 0.035 = 1.05\text{Nm}$$

Torque per motor:

$$T_{\text{motor}} \approx 1\text{Nm}$$